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BC230414205: SHEHR BANO

MTH622 - Vectors and Classical Mechanics (Quiz No 1)

Question # 1 of 10 (Start time: 10:39:59 PM, 23 November 2023)

The unit normal to the surface $x = 0$ is

Select the correct option

- | | |
|-----------------------|----------------|
| ☐ | does not exist |
| ☐ | $\pm \hat{j}$ |
| ☐ | $\pm \hat{i}$ |
| ☐ | $\pm \hat{k}$ |



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MTH622 - Vectors and Classical Mechanics (Quiz No 1)

Question # 2 of 10 (Start time: 10:40:41 PM, 23 November 2023)

If $\vec{\nabla} \cdot \vec{v} = 0$, then the vector field $\vec{v}$ is called

Select the correct option

☐	non solenoidal
☐	solenoidal

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MTH622 - Vectors and Classical Mechanics (Quiz No 1)

Question # 2 of 10 (Start time: 10:40:41 PM, 23 November 2023)

If $\vec{\nabla} \cdot \vec{v} = 0$, then the vector field $\vec{v}$ is called

Select the correct option

☐	non solenoidal
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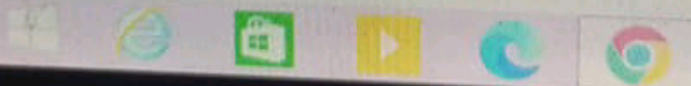
MTH622 - Vectors and Classical Mechanics (Quiz No 1)

Question # 3 of 10 (**Start time: 10:41:26 PM, 23 November 2023**)

If $\vec{v} = x\hat{i}$, then the vector field $\vec{v}$ is

Select the correct option

☐	non solenoidal
☐	solenoidal



MTH622 - Vectors and Classical Mechanics (Quiz No 1)

Question # 4 of 10 (Start time: 10:42:25 PM, 23 November 2023)

The unit normal to the surface $xy - 1 = 0$ is

Select the correct option



$$\frac{1}{\sqrt{x^2 + y^2}}(y\hat{i} + x\hat{j})$$



does not exist



$$\frac{1}{\sqrt{x^2 + y^2}}(y\hat{k} + x\hat{j})$$



$$\frac{1}{\sqrt{x^2 + y^2}}(y\hat{i} + x\hat{k})$$

To get future Google Chrome updates, you'll need Windows 10 or later. This computer is using

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MTH622 - Vectors and Classical Mechanics (Quiz No 1)

Question # 5 of 10 (**Start time: 10:43:15 PM, 23 November 2023**)

$\oint_C \vec{A} \cdot d\vec{r}$ for $\vec{A} = 2\hat{i}$ becomes

Select the correct option

☐ $2 \oint_C dx$

☐ $2 \oint_C \hat{i} dx$

To get future Google Chrome updates, you'll need Windows 10 or later. This computer is

BC230414205: SHEHR BANO

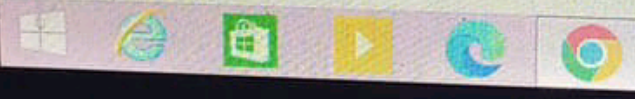
MTH622 - Vectors and Classical Mechanics (Quiz No 1)

Question # 6 of 10 (**Start time: 10:43:52 PM, 23 November 2023**)

A simple closed curve does not intersects itself anywhere.

Select the correct option

☐	false
☐	true



Question # 7 of 10 (Start time: 10:44:32 PM, 23 November 2023)

Unit circle is not a simple closed curve.

Select the correct option

☐	false
☐	true

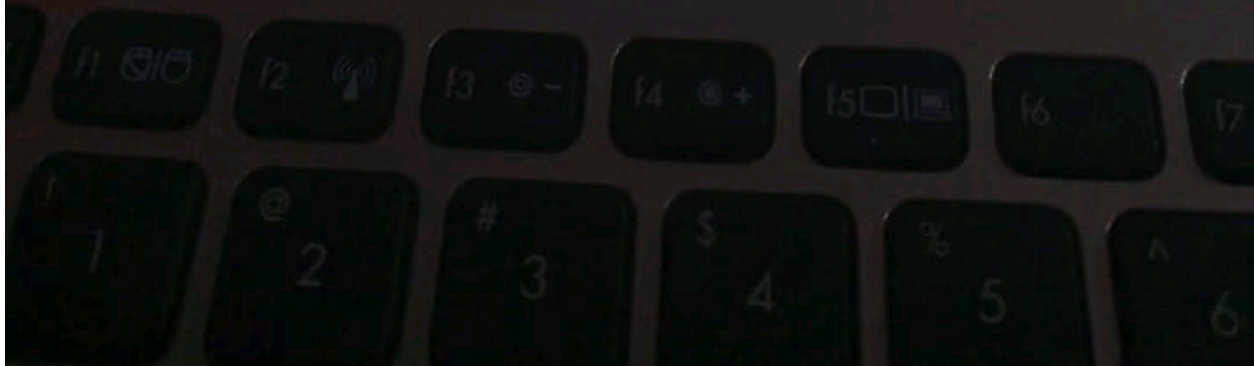


Question # 8 of 10 (Start time: 10:45:38 PM, 23 November 2023)

The Stoke's theorem uses which of the following operation?

Select the correct option

☐	Divergence
☐	Laplacian
☐	Curl
☐	Gradient



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BC230414205: SHEHR BANO

MTH622 - Vectors and Classical Mechanics (Quiz No 1)

Question # 9 of 10 (Start time: 10:46:33 PM, 23 November 2023)

If $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, then the integral is _____

Select the correct option

Independent



Dependent



BC230414205: SHEHR BANO

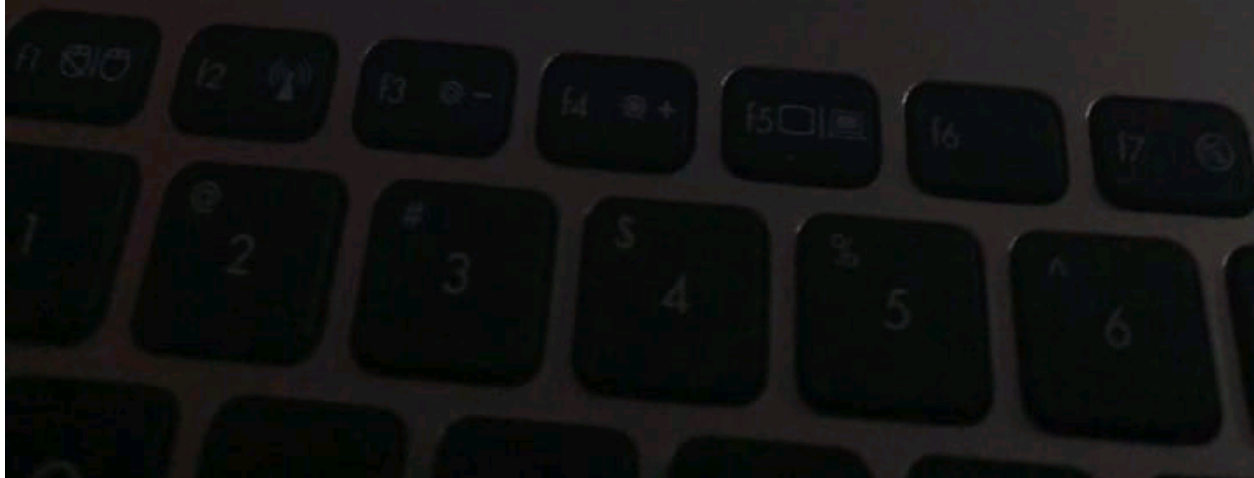
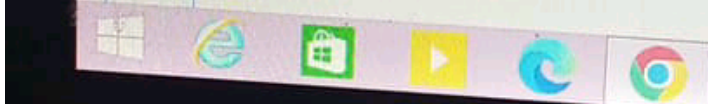
MTH622 - Vectors and Classical Mechanics (Quiz No 1)

Question # 10 of 10 (**Start time: 10:47:31 PM, 23 November 2023**)

The unit normal to the surface $z = 0$ is

Select the correct option

- | | |
|-----------------------|----------------|
| ☐ | does not exist |
| ☐ | $\pm \hat{j}$ |
| ☐ | $\pm \hat{k}$ |
| ☐ | $\pm \hat{i}$ |



Question # 1 of 10 (Start time: 10:53:51 PM, 23 November 2023)

The unit vector $\hat{j}$ is normal to the surface

Select the correct option

☐

$$y = 0$$

☐

none of these

☐

$$z = 0$$

☐

$$x = 0$$

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mth622

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F7

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BC230429387: SADAF IHSAN

MTH622 - Vectors and Classical Mechanics (Quiz No 1)

Question # 2 of 10 (Start time: 10:55:17 PM, 23 November 2023)

Both double and triple integrals can be used to calculate

Select the correct option

☐	volume of the given object
☐	None of these
☐	area under the curve
☐	length of curve

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MTH622 - Vectors and Classical Mechanics (Quiz No 1)

Question # 3 of 10 (Start time: 10:56:17 PM, 23 November 2023)

If $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$ then the integral is _____

Select the correct option

☐ Independent

☐ Dependent

MTH622 - Vectors and Classical Mechanics (Quiz No 1)

Question # 4 of 10 (**Start time: 10:57:11 PM, 23 November 2023**)

The second vector form of Green's theorem is also called

Select the correct option

tangential form of Green's theorem

☐

Normal form of Green's theorem

☐

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MTH622 - Vectors and Classical Mechanics (Quiz No 1)

Question # 5 of 10 (Start time: 10:58:04 PM, 23 November 2023)

 $\oint_C \vec{A} \cdot d\vec{r}$ for $\vec{A} = 2\hat{i}$ becomes

Select the correct option

☐ $2 \oint_C dx$

☐ $2 \oint_C \hat{i} dx$



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MTH622 - Vectors and Classical Mechanics (Quiz No 1)

Question # 6 of 10 (Start time: 10:58:41 PM, 23 November 2023)

For $\varphi = x$ the integration $\int_0^1 \int_0^1 \varphi \hat{k} dx dy$ gives

Select the correct option

☐ $\frac{1}{4} \hat{k}$

☐ $\frac{1}{2} \hat{k}$



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MTH622 - Vectors and Classical Mechanics (Quiz No 1)

Question # 7 of 10 (Start time: 10:59:13 PM, 23 November 2023)

Evaluate $\int \varphi d\vec{r}$ for $\varphi = 1$

Select the correct option

☐ $xyz(\hat{i} + \hat{j} + \hat{k})$

☐ $x\hat{i} + y\hat{j} + z\hat{k}$



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MTH622 - Vectors and Classical Mechanics (Quiz No 1)

Question # 8 of 10 (Start time: 10:59:57 PM, 23 November 2023)

$\hat{i} \times \vec{\nabla} \varphi$ for $\varphi = x$ becomes

Select the correct option

☐	$\vec{0}$
☐	$\hat{j}$

BC230429387: SADAF IHSAN

MTH622 - Vectors and Classical Mechanics (Quiz No 1)

Question # 9 of 10 (Start time: 11:01:22 PM, 23 November 2023)

If $\vec{v} = \hat{i}$, then the vector field $\vec{v}$ is

Select the correct option

☐ solenoidal

☐ non solenoidal



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$$\vec{\nabla} \psi \cdot d\vec{S} = \frac{\partial \psi}{\partial n} \hat{n}$$

$$\nabla\psi.\hat{n} = \frac{\partial\psi}{\partial n} \text{ and } \nabla\varphi.\hat{n} = \frac{\partial\varphi}{\partial n}$$

$$\nabla\psi.d\vec{S} = \nabla\psi.\hat{n}dS = \frac{\partial\psi}{\partial n}dS$$

23:10

Quiz

BC230426395: ANILA FATIMA

Time Left

84 sec(s)

MTH622 - Vectors and Classical Mechanics (Quiz No 1)

Quiz Start Time: 11:10 PM, 23 November 2023

Question # 1 of 10 (Start time: 11:10:46 PM, 23 November 2023)

Total Marks: 1

$\oint_C \vec{A} \cdot d\vec{r}$ for $\vec{A} = 2\hat{i}$ becomes

Select the correct option

Reload Math Equations

- ☐

$2 \oint_C dx$
- ☐

$2 \oint_C \hat{i} dx$

Click to Save Answer & Move to Next Question



Quiz

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BC230426395: ANILA FATIMA

Time Left 88
sec(s)

MTH622 - Vectors and Classical Mechanics (Quiz No 1)

Quiz Start Time: 11:10 PM, 23 November 2023

Question # 2 of 10 (start time: 11:11:25 PM, 23 November 2023)

Total Marks: 1

The unit vector $\hat{j}$ is normal to the surface

Select the correct option

 Reload Math Equations

☐	$x = 0$
☐	$z = 0$
☐	none of these
☐	$y = 0$

Click to Save Answer & Move to Next Question



Quiz

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BC230426395: ANILA FATIMA

Time Left 87
sec(s)

MTH622 - Vectors and Classical Mechanics (Quiz No 1)

Quiz Start Time: 11:10 PM, 23 November 2023

Question # 3 of 10 (Start time: 11:12:19 PM, 23 November 2023)

Total Marks: 1

If $\vec{v} = x\hat{i}$, then the vector field $\vec{v}$ is

Select the correct option

 Reload Math Equations☐ non solenoidal☐ solenoidal[Click to Save Answer & Move to Next Question](#)



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BC230426395: ANILA FATIMA

Time Left 88
sec(s)

MTH622 - Vectors and Classical Mechanics (Quiz No 1)

Quiz Start Time: 11:10 PM, 23 November 2023

Question # 4 of 10 (start time: 11:13:30 PM, 23 November 2023)

Total Marks: 1

The functions in Green's theorem will be

Select the correct option

 Reload Math Equations

☐	Discrete derivatives
☐	Continuous partial derivatives
☐	Discrete partial derivatives
☐	Continuous derivatives

Click to Save Answer & Move to Next Question



WhatsApp 23:14

MTH 622

~ asadj4026: b lgta ha

MTH622 - Vectors and Classical Mechanics (Quiz No 1)

Quiz Start Time: 11:10 PM, 23 November 2023

Question # 5 of 10 (Start time: 11:14:43 PM, 23 November 2023)

Total Marks: 1

The Green's theorem in plane can be written in tangential form as

Select the correct option



Reload Math Equations

☐ $\oint_C \vec{A} \cdot d\vec{r} = \int \int_R (\vec{\nabla} \times \vec{A}) \cdot \hat{k} dR$

☐ $\oint_C \vec{B} \cdot \hat{n} dS = \int \int_S \vec{\nabla} \cdot \vec{A} dR$

Click to Save Answer & Move to Next Question



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Time Left 87
sec(s)

MTH622 – Vectors and Classical Mechanics (Quiz No 1)

Quiz Start Time: 11:10 PM, 23 November 2023

Question # 6 of 10 (Start time: 11:15:44 PM, 23 November 2023)

Total Marks: 1

The Stoke's theorem uses which of the following operation?

Select the correct option

☐	Laplacian
☐	Divergence
☐	Gradient
☐	Curl

Click to Save Answer & Move to Next Question



Quiz

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BC230426395: ANILA FATIMA

Time Left 84
sec(s)

MTH622 - Vectors and Classical Mechanics (Quiz No 1)

Quiz Start Time: 11:10 PM, 23 November 2023

Question # 7 of 10 (Start time: 11:17:08 PM, 23 November 2023)

Total Marks: 1

The unit normal to the surface $xy - 1 = 0$ is

Select the correct option

 Reload Math Equations

- | | |
|-----------------------|---|
| ☐ | $\frac{1}{\sqrt{x^2+y^2}}(y\hat{i} + x\hat{j})$ |
| ☐ | does not exist |
| ☐ | $\frac{1}{\sqrt{x^2+y^2}}(y\hat{k} + x\hat{j})$ |
| ☐ | $\frac{1}{\sqrt{x^2+y^2}}(y\hat{i} + x\hat{k})$ |

Click to Save Answer & Move to Next Question



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BC230426395: ANILA FATIMA

Time Left 87
sec(s)

MTH622 - Vectors and Classical Mechanics (Quiz No 1)

Quiz Start Time: 11:10 PM, 23 November 2023

Question # 8 of 10 (Start time: 11:19:17 PM, 23 November 2023)

Total Marks: 1

 $\oint_C d\vec{r} \times \vec{B}$ for $\vec{B} = x\hat{i}$ becomes

Select the correct option

 Reload Math Equations☐ $\oint_C (xdz\hat{j} - xdy\hat{k})$ ☐ $\oint_C (xdz\hat{i} + xdy\hat{k})$

Click to Save Answer & Move to Next Question



Quiz

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BC230426395: ANILA FATIMA

Time Left 84
sec(s)

MTH622 - Vectors and Classical Mechanics (Quiz No 1)

Quiz Start Time: 11:10 PM, 23 November 2023

Question # 9 of 10 (Start time: 11:20:42 PM, 23 November 2023)

Total Marks: 1

The unit normal to the surface $z = 0$ is

Select the correct option

 Reload Math Equations

- | | |
|-----------------------|----------------|
| ☐ | $\pm \hat{j}$ |
| ☐ | does not exist |
| ☐ | $\pm \hat{k}$ |
| ☐ | $\pm \hat{i}$ |

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BC230426395: ANILA FATIMA

Time Left

81

sec(s)

MTH622 - Vectors and Classical Mechanics (Quiz No 1)

Quiz Start Time: 11:10 PM, 23 November 2023

Question # 10 of 10 (Start time: 11:21:36 PM, 23 November 2023)

Total Marks: 1

For $\varphi = x$ the integration $\int_0^1 \int_0^1 \varphi k dx dy$ gives

Select the correct option



Reload Math Equations

 $\frac{1}{4}k$  $\frac{1}{2}k$ 

Click to Save Answer & Move to Next Question

BC220422104: IQRA GHAFAR

MTH622 - Vectors and Classical Mechanics (Quiz No 1)

Question # 6 of 10 (Start time: 10:11:23 AM, 24 November 2023)

Gauss divergence theorem is used to derive equation of which phenomena

Select the correct option

- | | |
|-----------------------|------------------|
| ☐ | all of these |
| ☐ | flow of fluids |
| ☐ | heat conduction |
| ☐ | wave propagation |

BC220422104: IQRA GHAFAR

MTH622 - Vectors and Classical Mechanics (Quiz No 1)

Question # 9 of 10 (Start time: 10:13:15 AM, 24 November 2023)

$\oint_C d\vec{r} \times \vec{B}$ for $\vec{B} = x\hat{i}$ becomes

Select the correct option

☐ $\oint_C (xdz\hat{j} - xdy\hat{k})$

☐ $\oint_C (xdz\hat{i} + xdy\hat{k})$

BC220422104: IQRA GHAFAR

MTH622 - Vectors and Classical Mechanics (Quiz No 1)

Question # 8 of 10 (Start time: 10:12:34 AM, 24 November 2023)

If $\hat{n}$ is the unit normal to the surface S and ψ is a scalar point function having continuous partial derivatives then $\vec{\nabla}\psi \cdot \hat{n} = \frac{\partial\psi}{\partial n}$ implies

Select the correct option

☐ $\vec{\nabla}\psi \cdot d\vec{S} = \frac{\partial\psi}{\partial n} \hat{n}$

☐ $\vec{\nabla}\psi \cdot d\vec{S} = \frac{\partial\psi}{\partial n} dS$

BC220422104: IQRA GHAFAR

MTH622 - Vectors and Classical Mechanics (Quiz No 1)

Question # 7 of 10 (Start time: 10:12:01 AM, 24 November 2023)

If $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, then the integral is _____ of path.

Select the correct option

☐ Independent of path

☐ Dependent of path

If $\hat{n}$ is the unit normal to the surface S and ψ is a scalar point function having continuous partial derivatives then $\vec{\nabla} \psi \cdot \hat{n} = \frac{\partial \psi}{\partial n}$ implies

Select the correct option

 Reload Math Equations


$$\vec{\nabla} \psi \cdot d\vec{S} = \frac{\partial \psi}{\partial n} dS$$

☐
$$\vec{\nabla} \psi \cdot d\vec{S} = \frac{\partial \psi}{\partial n} \hat{n}$$

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MTH622 - Vectors and Classical Mechanics (Quiz No 1)

Question # 10 of 10 (Start time: 10:14:48 AM, 24 November 2023)

The unit normal to the surface $xy - 1 = 0$ is

Select the correct option

☐ $\frac{1}{\sqrt{x^2+y^2}}(y\hat{k} + x\hat{j})$

☐ $\frac{1}{\sqrt{x^2+y^2}}(y\hat{i} + x\hat{k})$

☐ does not exist

☐ $\frac{1}{\sqrt{x^2+y^2}}(y\hat{i} + x\hat{j})$



wave propagation



heat conduction

2

MTH622 – Vectors and Classical Mechanics (Quiz No 1)

Question # 3 of 10 (Start time: 02:15:24 AM, 24 November 2023)

For $\varphi = y$ the integration $\int_0^1 \int_0^1 \varphi \hat{k} dx dy$ gives

Select the correct option



$$\frac{1}{2} \hat{k}$$



$$\frac{1}{4} \hat{k}$$

Question # 2 of 10 (Start time: 11:00:41 AM, 24 November 2023)

The tangential form of Green's theorem in plane is also called Stokes theorem in the plane.

Select the correct option

☐	true
☐	false

Question # 9 of 10 (Start time: 11:03:46 AM, 24 November 2023)

The unit vector $\hat{i}$ is normal to the surface

Select the correct option

- | | |
|-----------------------|---------------|
| ☐ | none of these |
| ☐ | $x = 0$ |
| ☐ | $y = 0$ |
| ☐ | $z = 0$ |

MTH622 - Vectors and Classical Mechanics (Quiz No 1)

Question # 7 of 10 (Start time: 11:02:40 AM, 24 November 2023)

The unit normal to the xy plane is

Select the correct option

☐ $\pm \hat{k}$

☐ $\pm \hat{j}$

☐ $\pm \hat{i}$

☐ does not exist

Question # 3 of 10 (Start time: 11:01:05 AM, 24 November 2023)

If $\vec{v} = \hat{i}$, then the vector field $\vec{v}$ is

Select the correct option

☐	solenoidal
☒	non solenoidal



MTH622 - Vectors and Classical Mechanics (Quiz No 1)

Question # 2 of 10 (Start time: 11:33:49 AM, 24 November 2023)

The unit normal to the surface $z = 0$ is

Select the correct option

- | | |
|----------------------------------|----------------|
| ☐ | does not exist |
| ☐ | $\pm \hat{j}$ |
| ☐ | $\pm \hat{i}$ |
| ☒ | $\pm \hat{k}$ |

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Question # 5 of 10 (Start time: 11:36:54 AM, 24 November 2023)

If $\vec{\nabla} \cdot \vec{v} = 0$, then the vector field $\vec{v}$ is called

Select the correct option

☒	solenoidal
☐	non solenoidal

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MTH622 - Vectors and Classical Mechanics (Quiz No 1)

Question # 8 of 10 (Start time: 11:39:12 AM, 24 November 2023)

The functions in Green's theorem will be

Select the correct option

- | | |
|----------------------------------|--------------------------------|
| ☐ | Discrete partial derivatives |
| ☐ | Continuous derivatives |
| ☒ | Continuous partial derivatives |
| ☐ | Discrete derivatives |

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MTH622 – Vectors and Classical Mechanics (Quiz No 1)

Question # 10 of 10 (Start time: 11:41:27 AM, 24 November 2023)

The unit normal to the surface $x + y - 2 = 0$ is

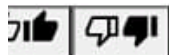
Select the correct option

- | | |
|----------------------------------|---|
| ☐ | $\frac{1}{\sqrt{2}}(\hat{i} + \hat{k})$ |
| ☐ | $\frac{1}{\sqrt{2}}(\hat{k} + \hat{j})$ |
| ☒ | $\frac{1}{\sqrt{2}}(\hat{i} + \hat{j})$ |
| ☐ | does not exist |

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Which relation satisfies the Green's theorem?

Select the correct option

Reload Math Equation

- ☐ $\oint_C (M dx + N dy) = \int_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dxdy$
- ☐ $\oint_C (M dx + N dy) = \int_R \left(\frac{\partial N}{\partial x} + \frac{\partial M}{\partial y} \right) dxdy$
- ☐ $\oint_C (M dx + N dy) = \int_R \left(\frac{\partial N}{\partial x} \right) dxdy + \int_R \left(\frac{\partial M}{\partial y} \right) dxdy$
- ☐ $\oint_C (M dx + N dy) = \int_R \left(\frac{\partial N}{\partial x} \right) dxdy + \int_R \left(\frac{\partial M}{\partial y} \right) dxdy$

Click to Save Answer & Move to Next Question



Question # 2 of 10 (start time: 11:42:54 AM, 24 November 2023)

A fluid of density $\rho(x, y, z, t)$ moves with velocity $v(x, y, z, t)$. If there are no sources or sinks then

Select the correct option

Reload

☐	$\frac{\partial \rho}{\partial t} = 0$
☐	$\nabla \cdot (\rho v) + \frac{\partial v}{\partial t} = 0$
☐	$\nabla \cdot (\rho v) = 0$
☒	$\nabla \cdot (\rho v) + \frac{\partial \rho}{\partial t} = 0$

Click to Save Answer & Move to



Question # 9 of 10 (Start time: 11:40:14 AM, 24 November 2023)

$\hat{i} \times \vec{\nabla} \varphi$ for $\varphi = x$ becomes

Select the correct option

$\hat{j}$



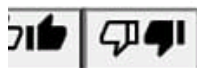
$\vec{0}$



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Question # 7 of 10 (Start time: 11:37:52 AM, 24 November 2023)

A fluid of density $\rho(x, y, z, t)$ moves with velocity $v(x, y, z, t)$. If there are no sources or sinks then

Select the correct option

☐	$\nabla \cdot (\rho v) = 0$
☒	$\frac{\partial \rho}{\partial t} = 0$
☐	$\nabla \cdot (\rho v) + \frac{\partial v}{\partial t} = 0$
☐	$\nabla \cdot (\rho v) + \frac{\partial \rho}{\partial t} = 0$

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Question # 5 of 10 (Start time: 11:44:36 AM, 24 November 2023)

The Stoke's theorem uses which of the following operation?

Select the correct option

- | | |
|-----------------------|------------|
| ☐ | Divergence |
| ☐ | Gradient |
| ☐ | Laplacian |
| ☐ | Curl |

Click to Save Answer &

Question # 1 of 10 (Start time: 11:26:14 AM, 24 November 2023)

For $\varphi = y$ the integration $\int_0^1 \int_0^1 \varphi \hat{k} dx dy$ gives



Select the correct option

☐ $\frac{1}{4} \hat{k}$

☐ $\frac{1}{2} \hat{k}$



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MTH622 – Vectors and Classical Mechanics (Quiz No 1)

Quiz

Question # 6 of 10 (Start time: 11:46:25 AM, 24 November 2023)

The functions in Green's theorem will be

Select the correct option

- | | |
|-----------------------|--------------------------------|
| ☐ | Discrete partial derivatives |
| ☐ | Discrete derivatives |
| ☐ | Continuous derivatives |
| ☐ | Continuous partial derivatives |

Click to S



MTH622 - Vectors and Classical Mechanics (Quiz No 1)

Quiz

Question # 7 of 10 (Start time: 11:47:00 AM, 24 November 2023)

 $\hat{j} \times \vec{\nabla} \varphi$ for $\varphi = j$ becomes

Select the correct option

 $\hat{i}$  $\vec{0}$ 

Click to S



The unit normal to the xy plane is

Select the correct option



- | | |
|-----------------------|----------------|
| ☐ | $\pm \hat{j}$ |
| ☐ | $\pm \hat{k}$ |
| ☐ | $\pm \hat{i}$ |
| ☐ | does not exist |

Click to Save Answer & Mo



11:36:04 AM, 24 November 2023

Question # 4 of 10 (Start time: 11:36:04 AM, 24 November 2023)

The Green's theorem in plane can be written in tangential form as

Select the correct option



$$\oint_C \vec{B} \cdot \hat{n} dS = \int \int_S \vec{\nabla} \cdot \vec{A} dR$$



$$\oint_C \vec{A} \cdot d\vec{r} = \int \int_R (\vec{\nabla} \times \vec{A}) \cdot \vec{k} dR$$



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**MTH622 – Vectors and Classical Mechanics (Quiz No 1)****Question # 9 of 10 (Start time: 11:49:26 AM, 24 November 2023)**

The second vector form of Green's theorem is also called

Select the correct option

tangential form of Green's theorem



Normal form of Green's theorem



Question # 10 of 10 (Start time: 11:50:12 AM, 24 November 2023)

The unit vector $\hat{j}$ is normal to the surface

Select the correct option

☐	$z = 0$
☐	$x = 0$
☒	$y = 0$
☐	none of these



MTH622 - Vectors and Classical Mechanics (Quiz No 1)

Question # 3 of 10 (Start time: 11:34:39 AM, 24 November 2023)

The divergence theorem converts

Select the correct option

- | | |
|-----------------------|----------------------------|
| ☐ | Surface to line integral |
| ☐ | Surface to volume integral |
| ☐ | Line to surface integral |
| ☐ | Volume to line integral |

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Question # 1 of 10 (Start time: 11:56:40 AM, 24 November 2023)

If $\hat{n}$ is the unit normal to the surface S and ψ is a scalar point function having continuous partial derivatives then $\vec{\nabla} \psi \cdot d\vec{S} = \frac{\partial \psi}{\partial n} dS$

Select the correct option



☒ $\vec{\nabla} \psi \cdot d\vec{S} = \frac{\partial \psi}{\partial n} dS$

☐ $\vec{\nabla} \psi \cdot d\vec{S} = \frac{\partial \psi}{\partial n} \hat{n}$

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The unit normal to the surface $xy - 1 = 0$ is

Select the correct option



$$\frac{1}{\sqrt{x^2+y^2}}(y\hat{i} + x\hat{k})$$



does not exist



$$\frac{1}{\sqrt{x^2+y^2}}(y\hat{i} + x\hat{j})$$



$$\frac{1}{\sqrt{x^2+y^2}}(y\hat{k} + x\hat{j})$$

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Question # 8 of 10 (Start time: 02:48:30 AM, 24 November 2023)

Total Marks: 1

$\oint_C d\vec{r} \times \vec{B}$ for $\vec{B} = y\hat{j}$ becomes

Select the correct option

 Reload Math Equations

☐ $\oint_C (-ydz\hat{i} + ydx\hat{k})$



☒ $\oint_C (ydz\hat{i} + ydy\hat{k})$



Question # 4 of 10 (Start time: 12:01:29 PM, 24 November 2020)

The identity $\oint_C \vec{B} \times d\vec{r} = \int \int_S (\hat{n} \times \vec{\nabla}) \times \vec{B} dS$ is

Select the correct option

false



true



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Question # 5 of 10 (Start time: 12:02:02 PM, 24 November 2023)

Unit circle is not a simple closed curve.

Select the correct option

true



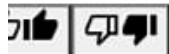
false



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Question # 7 of 10 (Start time: 12:03:53 PM, 24 November 2023)

The tangential form of Green's theorem in plane is also called Stokes theorem in the plane.

Select the correct option

☐	true
☐	false

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Question # 9 of 10 (Start time: 12:06:46 PM, 24 November 2023)

$\oint_C d\vec{r} \times \vec{B}$ for $\vec{B} = y\hat{j}$ becomes

Select the correct option

$\oint_C (-ydz\hat{i} + ydx\hat{k})$



$\oint_C (ydz\hat{i} + ydy\hat{k})$



**Question # 10 of 10 (Start time: 12:09:40 PM, 24 November 2023)**

The second vector form of Green's theorem is also called

Select the correct option

☐	tangential form of Green's theorem
☒	Normal form of Green's theorem

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Question # 2 of 10 (Start time: 12:12:16 PM, 24 November 2023)

The unit normal to the surface $x = 0$ is

Select the correct option

- | | |
|----------------------------------|----------------|
| ☐ | does not exist |
| ☒ | $\pm \hat{j}$ |
| ☐ | $\pm \hat{k}$ |
| ☐ | $\pm \hat{i}$ |



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Time Left

87

sec(s)

MTH522 - Vectors and Classical Mechanics (Quiz No 1)

Quiz Start Time: 12:12 PM, 24 November 2023

Question # 1 of 10 (Start time: 12:12:57 PM, 24 November 2023)

Total Marks: 1

Evaluate $\int \varphi d\vec{r}$ for $\varphi = 1$

Select the correct option

Reload Math Equations

- | | |
|-----------------------|------------------------------------|
| ☐ | $xyz(\hat{i} + \hat{j} + \hat{k})$ |
| ☐ | $z\hat{i} + y\hat{j} + z\hat{k}$ |

Click to Save Answer & Move to Next Question

Gauss divergence theorem is used to derive equation of which phenomena

Select the correct option

- | | |
|-----------------------|------------------|
| ☐ | flow of fluids |
| ☐ | heat conduction |
| ☐ | wave propagation |
| ☐ | all of these |

The identity $\oint_C \vec{B} \times d\vec{r} = \int \int_S (\hat{n} \times \vec{\nabla}) \times \vec{B} dS$ is

Select the correct option

☐	false
☐	true



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Time Left 88 sec(s)

MTH522 - Vectors and Classical Mechanics (Quiz No 1)

Quiz Start Time: 12:12 PM, 24 November 2023

Question # 2 of 10 (Start time: 12:13:46 PM, 24 November 2023)

Total Marks: 1

The tangential form of Green's theorem in plane is also called Stokes theorem in the plane.

Select the correct option

Reload Math Equations

☐	true
☐	false

Click to Save Answer & Move to Next Question



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Time Left

88

sec(s)

MTH522 - Vectors and Classical Mechanics (Quiz No 1)

Quiz Start Time: 12:12 PM, 24 November 2023

Question # 3 of 10 (Start time: 12:14:30 PM, 24 November 2023)

Total Marks: 1

The unit normal to the surface $x + y - 2 = 0$ is

Select the correct option

Reload Math Equations

- ☐ $\frac{1}{\sqrt{2}}(\vec{i} + \vec{j})$
- ☐ does not exist
- ☐ $\frac{1}{\sqrt{2}}(\vec{k} + \vec{j})$
- ☐ $\frac{1}{\sqrt{2}}(\vec{i} + \vec{k})$

Click to Save Answer & Move to Next Question

Question # 5 of 10 (Start time: 12:14:41 PM, 24 November 2023)

The unit vector $\hat{i}$ is normal to the surface

Select the correct option

- | | |
|----------------------------------|---------------|
| ☐ | none of these |
| ☒ | $x = 0$ |
| ☐ | $y = 0$ |
| ☐ | $z = 0$ |

Question # 6 of 10 (Start time: 12:15:12 PM, 24 November 2023)

The unit normal to the surface $z = 0$ is

Select the correct option

☐

$\pm \hat{i}$

☐

does not exist

☐

$\pm \hat{j}$

☐

$\pm \hat{k}$



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Time Left

87

sec(s)

MTH522 - Vectors and Classical Mechanics (Quiz No 1)

Quiz Start Time: 12:12 PM, 24 November 2023

Question # 4 of 10 (Start time: 12:15:26 PM, 24 November 2023)

Total Marks: 1

The unit normal to the surface $xy - 1 = 0$ is

Select the correct option

Reload Math Equations

- ☐ $\frac{1}{\sqrt{x^2+y^2}}(y\hat{i} + x\hat{k})$
- ☐ $\frac{1}{\sqrt{x^2+y^2}}(y\hat{k} + x\hat{j})$
- ☐ does not exist
- ☐ $\frac{1}{\sqrt{x^2+y^2}}(y\hat{i} + x\hat{j})$

Click to Save Answer & Move to Next Question

Question # 7 of 10 (Start time: 12:16:15 PM, 24 November 2023)

$\hat{j} \times \vec{\nabla} \varphi$ for $\varphi = j$ becomes

Select the correct option

☐ $\hat{z}$

☐ $\vec{0}$

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MTH622 - Vectors and Classical Mechanics (Quiz No 1)

Question # 5 of 10 (Start time: 12:16:18 PM, 24 November 2023)

Both double and triple integrals can be used to calculate

Select the correct option

☐	None of these
☐	area under the curve
☐	volume of the given object
☐	length of curve



Type here to search



Question 4 of 10 (Start time: 12:17:26 PM, 24 November 2023)

The unit normal to the surface $yz - 1 = 0$ is

Select the correct option

- | | |
|-----------------------|---|
| ☐ | $\frac{1}{\sqrt{y^2+z^2}}(z\hat{j} + y\hat{k})$ |
| ☐ | does not exist |
| ☐ | $\frac{1}{\sqrt{y^2+z^2}}(y\hat{j} + z\hat{k})$ |
| ☐ | $\frac{1}{\sqrt{y^2+z^2}}(y\hat{i} + z\hat{k})$ |



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Time Left 86 sec(s)

MTH522 - Vectors and Classical Mechanics (Quiz No 1)

Quiz Start Time: 12:12 PM, 24 November 2023

Question # 5 of 10 (Start time: 12:17:59 PM, 24 November 2023)

Total Marks: 1

The unit vector $\hat{j}$ is normal to the surface

Select the correct option

Reload Math Equations

- | | |
|-----------------------|---------------|
| ☐ | $y = 0$ |
| ☐ | $z = 0$ |
| ☐ | none of these |
| ☐ | $x = 0$ |

Click to Save Answer & Move to Next Question

Question # 9 of 10 (Start time: 12:18:12 PM, 24 November 2023)

$\oint_C d\vec{r} \times \vec{B}$ for $\vec{B} = x\hat{i}$ becomes

Select the correct option

☐ $\oint_C (xdz\hat{j} - xdy\hat{k})$

☐ $\oint_C (xdz\hat{i} + xdy\hat{k})$

Question # 7 of 10 (Start time: 12:17:58 PM, 24 November 2023)

The unit normal to the surface $yz - 1 = 0$ is

▶ Select the correct option

☐ $\frac{1}{\sqrt{y^2+z^2}}(y\hat{j} + z\hat{k})$

☐ $\frac{1}{\sqrt{y^2+z^2}}(y\hat{i} + z\hat{k})$

☐ does not exist

☐ $\frac{1}{\sqrt{y^2+z^2}}(y\hat{j} + z\hat{i})$



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Time Left

87

sec(s)

MTH522 - Vectors and Classical Mechanics (Quiz No 1)

Quiz Start Time: 12:12 PM, 24 November 2023

Question # 6 of 10 (Start time: 12:18:48 PM, 24 November 2023)

Total Marks: 1

For $\varphi = x$ the integration $\int_0^1 \int_0^1 \varphi \hat{k} dx dy$ gives

Select the correct option

Reload Math Equations

☐ $\frac{1}{4} \hat{k}$

☐ $\frac{1}{2} \hat{k}$

Click to Save Answer & Move to Next Question

If $\vec{v} = x\hat{i}$, then the vector field $\vec{v}$ is

Select the correct option

☐ solenoidal

☐ non solenoidal



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Time Left

87

sec(s)

MTH522 - Vectors and Classical Mechanics (Quiz No 1)

Quiz Start Time: 12:12 PM, 24 November 2023

Question # 7 of 10 (Start time: 12:19:35 PM, 24 November 2023)

Total Marks: 1

The unit normal to the surface $x + z - 1 = 0$ is

Select the correct option

Reload Math Equations

- | | |
|-----------------------|---|
| ☐ | does not exist |
| ☐ | $\frac{1}{\sqrt{2}}(\hat{i} + \hat{j})$ |
| ☐ | $\frac{1}{\sqrt{2}}(\hat{i} + \hat{k})$ |
| ☐ | $\frac{1}{\sqrt{2}}(\hat{k} + \hat{j})$ |

Click to Save Answer & Move to Next Question



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Time Left 87 sec(s)

MTH522 - Vectors and Classical Mechanics (Quiz No 1)

Quiz Start Time: 12:12 PM, 24 November 2023

Question # 8 of 10 (Start time: 12:20:26 PM, 24 November 2023)

Total Marks: 1

The first vector form of Green's theorem is also called

Select the correct option

☐	Normal form of Green's theorem
☐	tangential form of Green's theorem

Click to Save Answer & Move to Next Question



where $dR = dxdy$

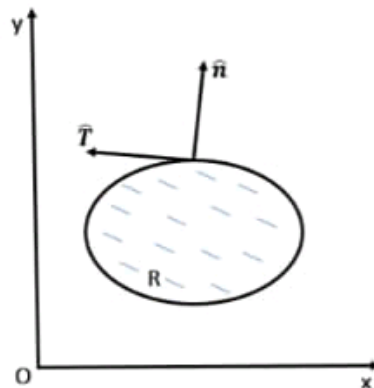
A generalization of this surface to S in space having C as boundary leads quite naturally to Stokes' theorem. This form of Green's theorem is sometimes called Stokes' theorem in the plane.

The Green's theorem in plane is a special case of Stokes theorem.

Second Vector Form (or normal form) of Green's Theorem

As we derive in first vector form of Green's theorem

$$Mdx + Ndy = \vec{A} \cdot d\vec{r} = \vec{A} \cdot \hat{T} ds$$



where $\hat{T} = \frac{d\vec{r}}{ds}$ = unit tangent vector to C as shown in figure.

If $\hat{n}$ is the outward drawn unit normal to C , then $\hat{T} = \hat{k} \times \hat{n}$ 47 / 63

$$Mdx + Ndy = \vec{A} \cdot \hat{T} ds = \vec{A} \cdot (\hat{k} \times \hat{n}) ds = (\vec{A} \times \hat{n}) \cdot \hat{k} ds$$

Since $\vec{A} = M\hat{i} + N\hat{j}$, therefore

$$\vec{B} = \vec{A} \times \hat{k} = (M\hat{i} + N\hat{j}) \times \hat{k} = N\hat{i} - M\hat{j}$$

and

$$\nabla \cdot \vec{B} = \frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}$$

then the equation (1) becomes

$$\oint_C \vec{B} \cdot \hat{n} ds = \iint_S \nabla \cdot \vec{B} dR$$

where $dR = dxdy$

these are the required vector notations of Green's theorem.



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Time Left

87
sec(s)

MTH522 - Vectors and Classical Mechanics (Quiz No 1)

Quiz Start Time: 12:12 PM, 24 November 2023

Question # 9 of 10 (Start time: 12:21:19 PM, 24 November 2023)

Total Marks: 1

 $\oint_C d\vec{r} \times \vec{B}$ for $\vec{B} = x\hat{i}$ becomes

Select the correct option

Reload Math Equations

$$\oint_C (xdz\hat{i} + xdy\hat{k})$$



$$\oint_C (xdz\hat{j} - xdy\hat{k})$$



Click to Save Answer & Move to Next Question



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82
sec(s)

MTH522 - Vectors and Classical Mechanics (Quiz No 1)

Quiz Start Time: 12:12 PM, 24 November 2023

Question # 10 of 10 (Start time: 12:23:04 PM, 24 November 2023)

Total Marks: 1

Gauss divergence theorem is used to derive equation of which phenomena

Select the correct option

Reload Math Equations

- | | |
|----------------------------------|------------------|
| ☐ | wave propagation |
| ☒ | all of these |
| ☐ | heat conduction |
| ☐ | flow of fluids |

Click to Save Answer & Move to Next Question

MTH622 – Vectors and Classical Mechanics (Quiz No 1)

Question # 3 of 10 (**Start time: 02:15:24 AM, 24 November 2023**)

For $\varphi = y$ the integration $\int_0^1 \int_0^1 \varphi \hat{k} dx dy$ gives

Select the correct option

☒ $\frac{1}{2} \hat{k}$

☐ $\frac{1}{4} \hat{k}$